Radar range and velocity

antocciphysics | TEACHER KEY | CPA

Learning target: Separate a radar’s range measurement from its line-of-sight velocity measurement.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 12; 3 course days later.*
Same speed c; the higher-frequency wavelength is 1/3 as large.

02. Explain a worked example (worked)
Total path 600 m; one-way range 300 m.

03. Look and explain (model)
Delay for range; Doppler frequency shift for radial velocity.

04. Build the solution (practice)
60 m total; 30 m range.

05. Use the idea (reason)
No; could move transversely to the line of sight.

06. Choose a method (mixed) *Review: Unit 6, Day 3; 12 course days later.*
35 m/s from $v=f\lambda$.

07. Exit check (exit)
500 m.

08. STRETCH: Apply it another way (stretch)
600 to 450 m; approaching, assuming consistent geometry and timing.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 12; 3 course days later.*
$\lambda \approx 0.06$ m; microwave.

02. Explain a worked example (worked)
Total path 600 m; one-way range 300 m.

03. Represent and interpret (model)
Delay for range; Doppler frequency shift for radial velocity.

04. Solve and justify (practice)
Range 6×10^6 m; radial speed 20 m/s.

05. Analyze the reasoning (reason)
No; could move transversely to the line of sight.

06. Choose a method (mixed) *Review: Unit 6, Day 3; 12 course days later.*
$v = 30$ m/s; $\lambda = 30/60 \approx 0.5$ m.

07. Exit check (exit)
Range describes position along line of sight; Doppler shift describes radial velocity.

08. STRETCH: Apply it another way (stretch)
-120 m/s; transverse motion is not determined by range change alone.